

#File Backup Logger

Description

A program that backs up files and directories with versioning, logs the operations locally, and includes a GUI.

Prepared for: Jesus Alamo, EuTech Engineers **Developer:** Gvantsa Lukhava **Architecture:** OOP Python
(Terminal testing + TKinter Gui Wrapper)

System Architecture & Code Features

Features

- **Flexible Backup:** Choose between raw directory copies or compressed `.zip` archives.
- **Detailed Logging:** Every operation is tracked in `backup_log.txt` with timestamps, file counts, and backup duration.
- **Error Resilient:** Built-in handling for permission errors and missing directories.
- **Persistence:** Remembers your last-used source folder and compression preference using `config.json`.
- **Modern Interface:** Built with `tkinter` for a simple, native desktop experience.

Prerequisites

- **Python 3.x**
- No external libraries required (uses standard library modules: `tkinter`, `pathlib`, `shutil`, `json`, `datetime`, `time`).

Project Structure

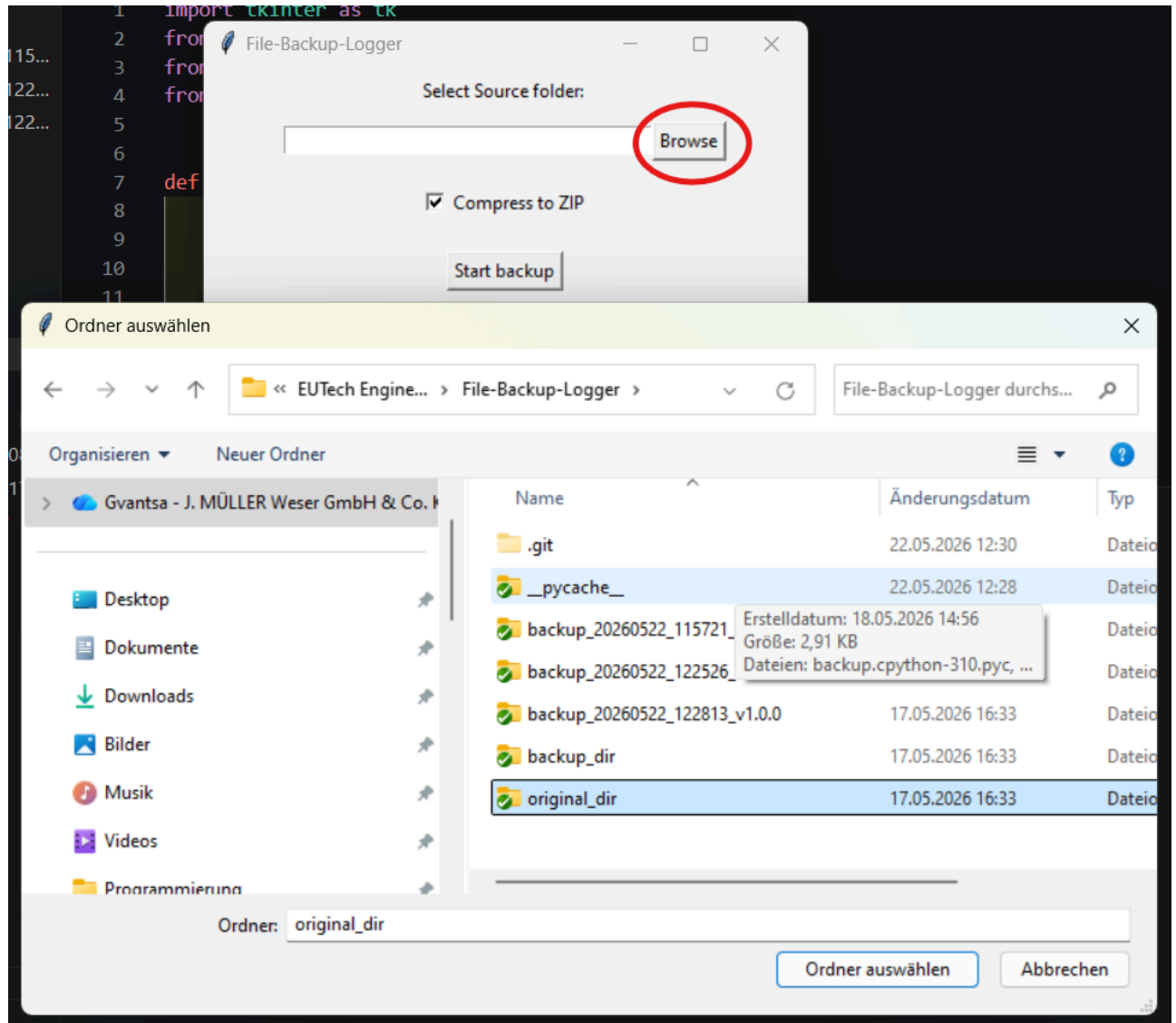
- `backup.py`: The core engine handling file system operations and logging.
- `gui.py`: The user interface layer.
- `config_manager.py`: Handles saving and loading user preferences.

How to Use

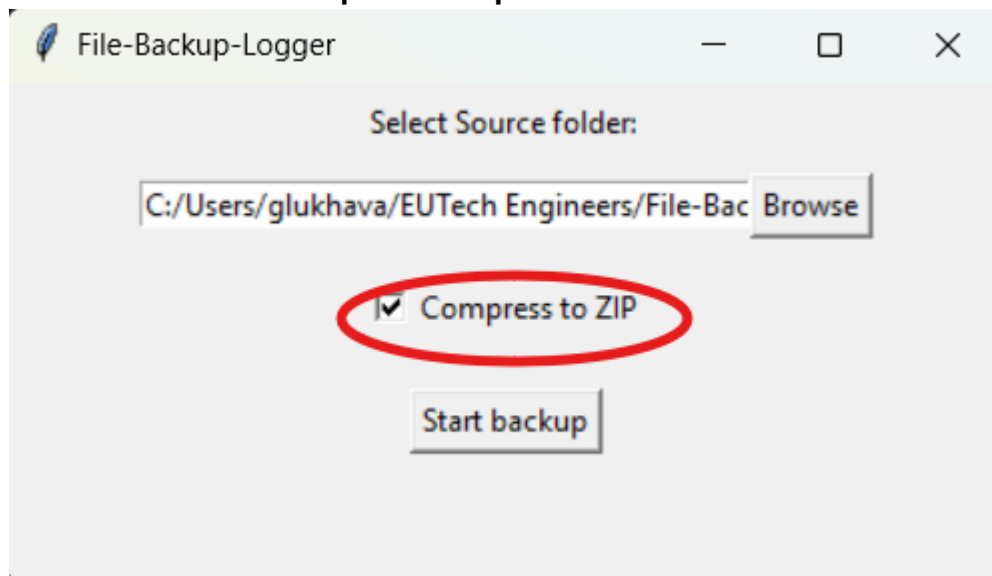
- Launch the application by running `python gui.py`

GUI

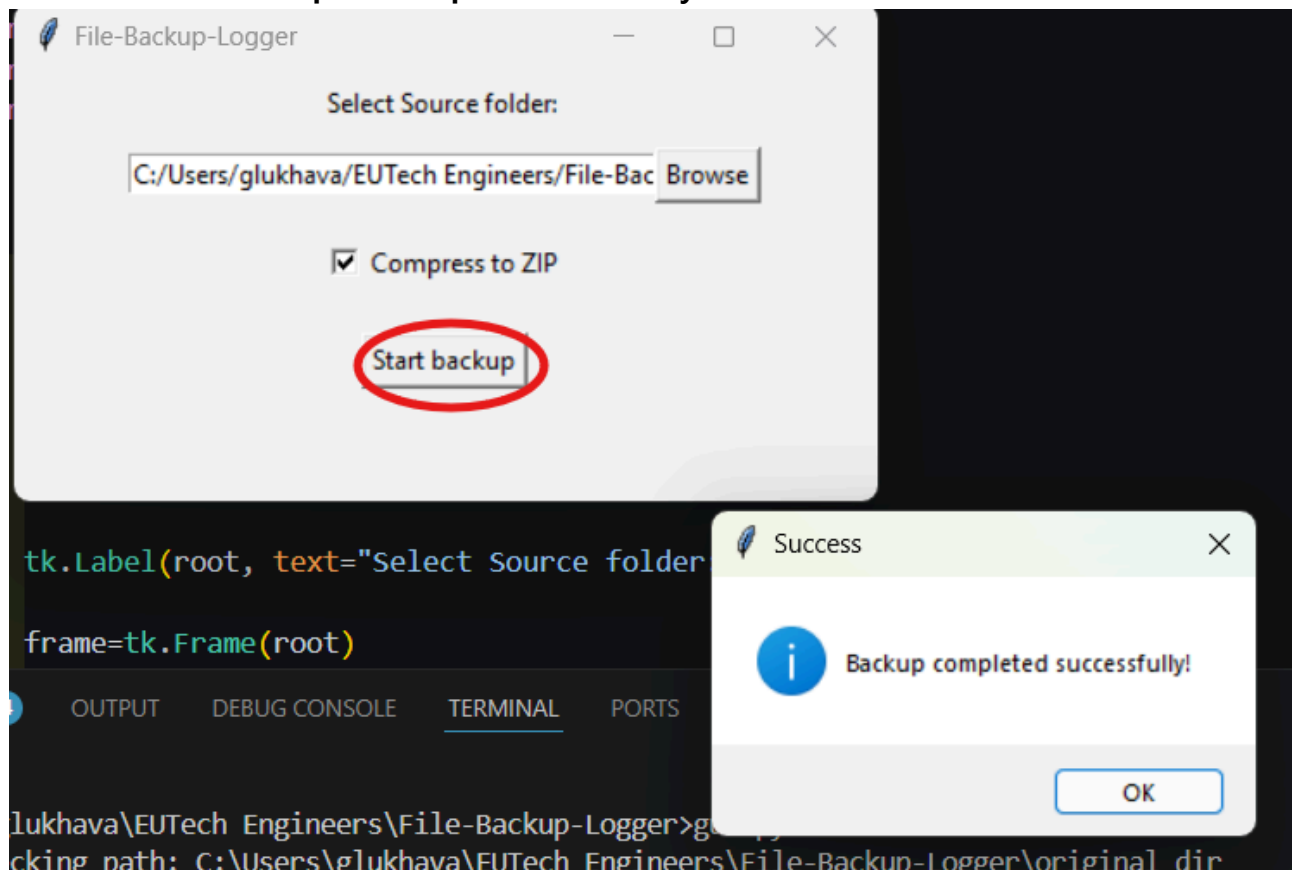
1. STEP 1: Browse into files



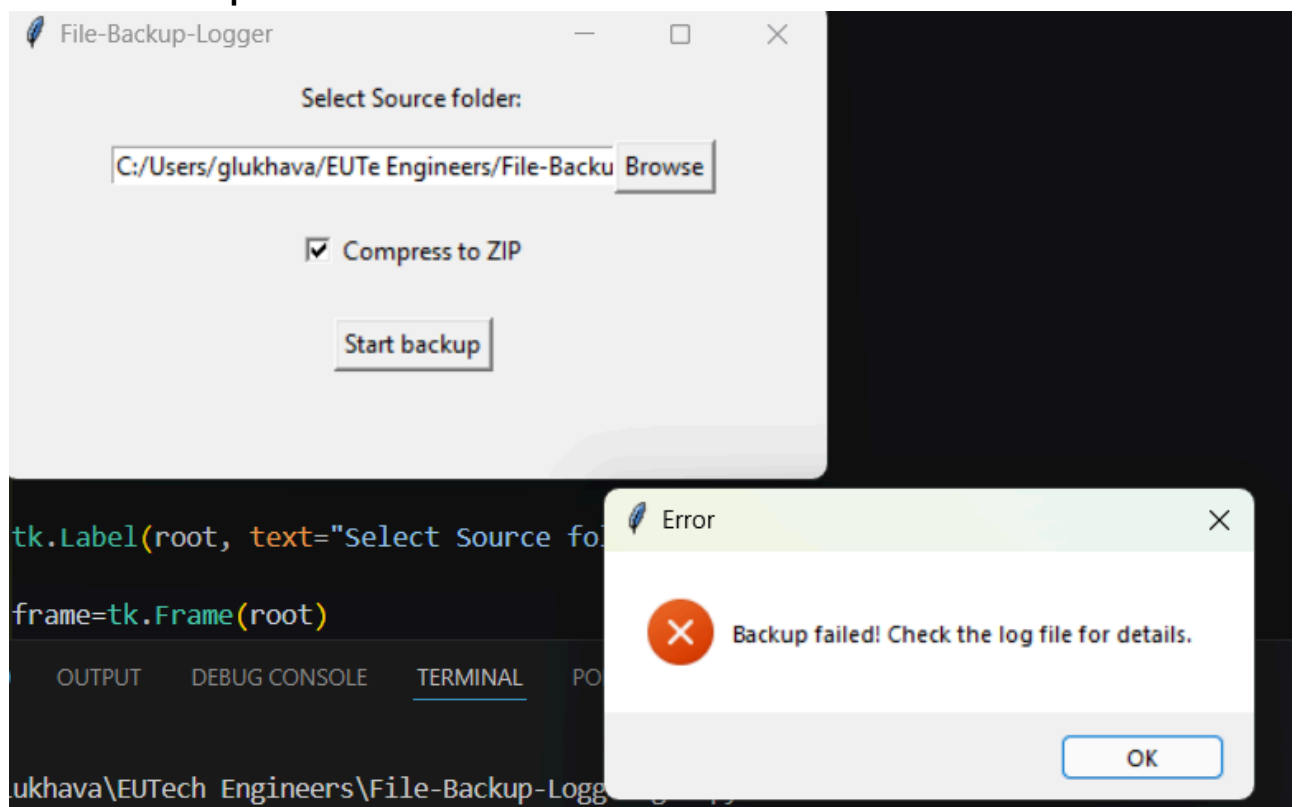
2. STEP 2: Select either to zip the backup or not:



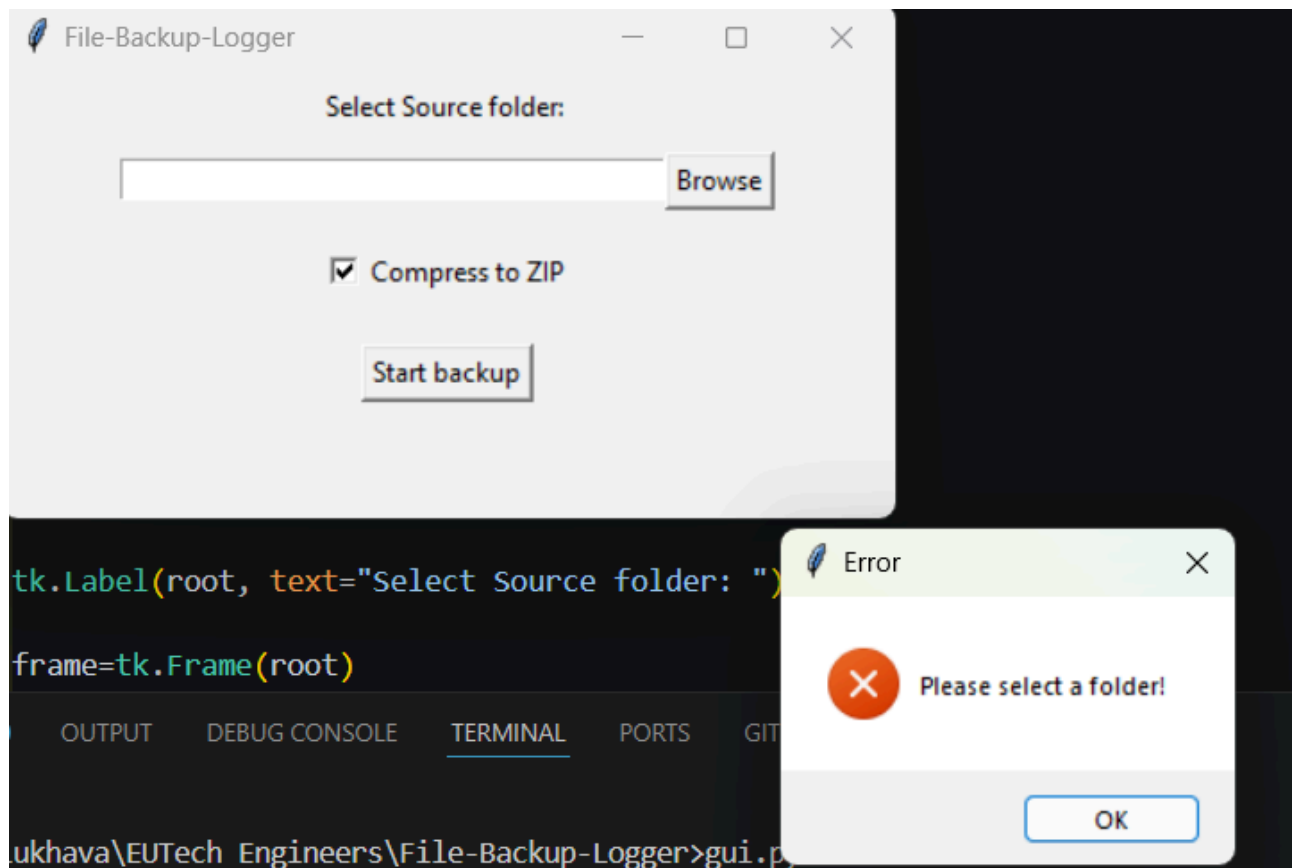
3. STEP 3: Start the backup and complete successfully:



4. STEP 4: To Backup file doesnt exist:



5. STEP 5: folder not selected :



CODE

1. Class Backup

```

backup.py M X  gui.py  backup_log.txt M  {} config.json M  README.md M  config_
File-Backup-Logger > backup.py > Backup
5  class Backup:
6
7      @staticmethod
8      def count_files(source_path):
9          return len([f for f in Path(source_path).rglob('*') if f.is_file()])
10
11     @staticmethod
12     def backup(backup_source, zip_it=False):
13         start_time=time.time()
14         VERSION="v1.0.0"
15         backup_source=Path(backup_source)
16         print(f"DEBUG: Checking path: {backup_source.absolute()}") # Add this!
17         # backup_source=Path(input("Enter source folder path: "))
18         timestamp = datetime.datetime.now().strftime("%Y%m%d_%H%M%S")
19         backup_destination=f'backup_{timestamp}_{VERSION}'
20
21         try:
22             if backup_source.exists() and backup_source.is_dir():
23                 if zip_it==True:
24                     zip_name=f'zip_{timestamp}_{VERSION}'
25                     shutil.make_archive(zip_name,format='zip',root_dir=str(backup_source))
26                     actual_destination=f"{zip_name}.zip"
27                     return True
28                 else:
29                     shutil.copytree(backup_source,backup_destination)
30                     actual_destination=backup_destination
31
32                     duration=time.time()-start_time
33                     file_count=Backup.count_files(backup_source)
34                     log_entry = (
35                         f"[{timestamp}] SUCCESS! | "
36                         f"Files: {file_count} | "
37                         f"Duration: {duration:.2f}s | "

```

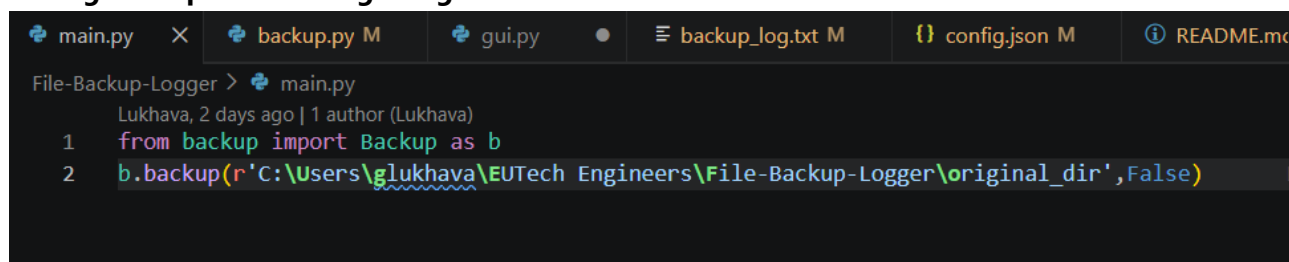
1.1

```

38         f"Source: {backup_source} | "
39         f"Dest: {actual_destination}\n"
40     )
41     with open("backup_log.txt","a") as log_file:
42         log_file.write(log_entry)
43     return True
44
45     else:
46         print("ERROR: That folder does not exist! ")
47         log_entry=f"[{timestamp}] ERROR! Could NOT be Backed up: {backup_source} As: {backup_destination}\n"
48         with open("backup_log.txt","a") as log_file:
49
50
51             log_file.write(log_entry)
52         return False
53     except PermissionError:
54         log_entry=f"[{timestamp}] Permission denied! Could NOT be Backed up: {backup_source} As: {backup_destination}\n"
55         with open("backup_log.txt","a") as log_file:
56             log_file.write(log_entry)
57         return False
58     except Exception as e:
59         log_entry=f"[{timestamp}] Error occurred! Could NOT be Backed up: {backup_source} As: {backup_destination}\n"
60         with open("backup_log.txt","a") as log_file:
61             log_file.write(log_entry)
62         print(f"Error; {e}")
63         return False

```

2. testing backup before integrating to GUI

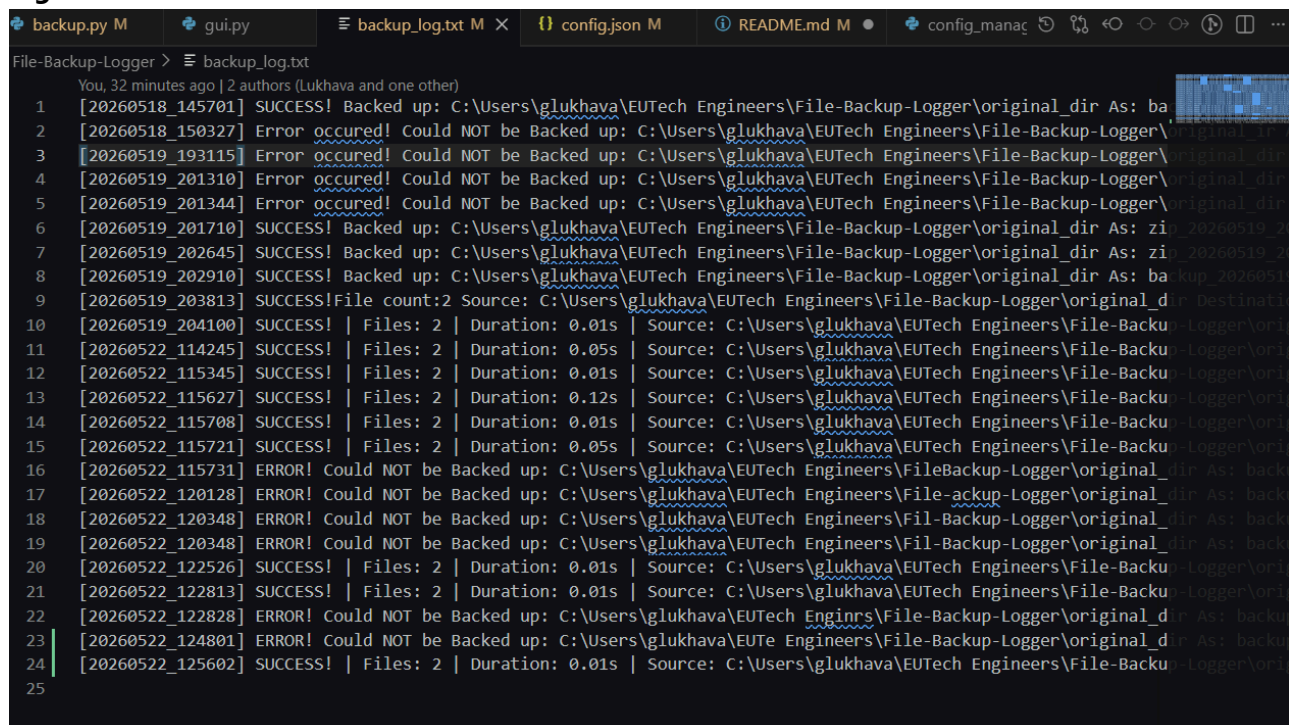


```

main.py × backup.py M gui.py backup_log.txt M config.json M README.md
File-Backup-Logger > main.py
Lukhava, 2 days ago | 1 author (Lukhava)
1 from backup import Backup as b
2 b.backup(r'C:\Users\glukhava\EUTech Engineers\File-Backup-Logger\original_dir',False)

```

3. log files:



```

backup.py M gui.py backup_log.txt M config.json M README.md M config_manag
File-Backup-Logger > backup_log.txt
You, 32 minutes ago | 2 authors (Lukhava and one other)
1 [20260518_145701] SUCCESS! Backed up: C:\Users\glukhava\EUTech Engineers\File-Backup-Logger\original_dir As: ba
2 [20260518_150327] Error occured! Could NOT be Backed up: C:\Users\glukhava\EUTech Engineers\File-Backup-Logger\original_dir
3 [20260519_193115] Error occured! Could NOT be Backed up: C:\Users\glukhava\EUTech Engineers\File-Backup-Logger\original_dir
4 [20260519_201310] Error occured! Could NOT be Backed up: C:\Users\glukhava\EUTech Engineers\File-Backup-Logger\original_dir
5 [20260519_201344] Error occured! Could NOT be Backed up: C:\Users\glukhava\EUTech Engineers\File-Backup-Logger\original_dir
6 [20260519_201710] SUCCESS! Backed up: C:\Users\glukhava\EUTech Engineers\File-Backup-Logger\original_dir As: zip_20260519_2
7 [20260519_202645] SUCCESS! Backed up: C:\Users\glukhava\EUTech Engineers\File-Backup-Logger\original_dir As: zip_20260519_2
8 [20260519_202910] SUCCESS! Backed up: C:\Users\glukhava\EUTech Engineers\File-Backup-Logger\original_dir As: backup_2026051
9 [20260519_203813] SUCCESS!File count:2 Source: C:\Users\glukhava\EUTech Engineers\File-Backup-Logger\original_dir Destinati
10 [20260519_204100] SUCCESS! | Files: 2 | Duration: 0.01s | Source: C:\Users\glukhava\EUTech Engineers\File-Backup-Logger\ori
11 [20260522_114245] SUCCESS! | Files: 2 | Duration: 0.05s | Source: C:\Users\glukhava\EUTech Engineers\File-Backup-Logger\ori
12 [20260522_115345] SUCCESS! | Files: 2 | Duration: 0.01s | Source: C:\Users\glukhava\EUTech Engineers\File-Backup-Logger\ori
13 [20260522_115627] SUCCESS! | Files: 2 | Duration: 0.12s | Source: C:\Users\glukhava\EUTech Engineers\File-Backup-Logger\ori
14 [20260522_115708] SUCCESS! | Files: 2 | Duration: 0.01s | Source: C:\Users\glukhava\EUTech Engineers\File-Backup-Logger\ori
15 [20260522_115721] SUCCESS! | Files: 2 | Duration: 0.05s | Source: C:\Users\glukhava\EUTech Engineers\File-Backup-Logger\ori
16 [20260522_115731] ERROR! Could NOT be Backed up: C:\Users\glukhava\EUTech Engineers\FileBackup-Logger\original_dir As: back
17 [20260522_120128] ERROR! Could NOT be Backed up: C:\Users\glukhava\EUTech Engineers\File-ackup-Logger\original_dir As: back
18 [20260522_120348] ERROR! Could NOT be Backed up: C:\Users\glukhava\EUTech Engineers\Fil-Backup-Logger\original_dir As: back
19 [20260522_120348] ERROR! Could NOT be Backed up: C:\Users\glukhava\EUTech Engineers\Fil-Backup-Logger\original_dir As: back
20 [20260522_122526] SUCCESS! | Files: 2 | Duration: 0.01s | Source: C:\Users\glukhava\EUTech Engineers\File-Backup-Logger\ori
21 [20260522_122813] SUCCESS! | Files: 2 | Duration: 0.01s | Source: C:\Users\glukhava\EUTech Engineers\File-Backup-Logger\ori
22 [20260522_122828] ERROR! Could NOT be Backed up: C:\Users\glukhava\EUTech Enginrs\File-Backup-Logger\original_dir As: backe
23 [20260522_124801] ERROR! Could NOT be Backed up: C:\Users\glukhava\EUTE Engineers\File-Backup-Logger\original_dir As: backu
24 [20260522_125602] SUCCESS! | Files: 2 | Duration: 0.01s | Source: C:\Users\glukhava\EUTech Engineers\File-Backup-Logger\ori
25

```

4. GUI()

```

main.py backup.py M gui.py backup_log.txt M config.json M README.m
File-Backup-Logger > gui.py > start_gui > run_backup
You, 43 minutes ago | 2 authors (Lukhava and one other)
1 import tkinter as tk
2 from tkinter import filedialog, messagebox
3 from backup import Backup
4 from config_manager import load_config, save_config
5
6 def start_gui():
7     root=tk.Tk()
8     root.title("File-Backup-Logger")
9     root.geometry("400x200")
10
11     path_var=tk.StringVar()
12     zip_var=tk.BooleanVar()
13
14     tk.Label(root, text="Select Source folder: ").pack(pady=5)
15
16     frame=tk.Frame(root)
17     frame.pack(pady=5)
18
19     tk.Entry(frame, textvariable=path_var,width=40).pack(side=tk.LEFT)
20     tk.Button(frame, text="Browse", command=lambda: path_var.set(filedialog.askdirectory())).pack(side=tk.LEFT)
21
22     tk.Checkbutton(root, text="Compress to ZIP", variable=zip_var).pack(pady=10)
23
24     config=load_config()
25     path_var.set(config.get("last_source",""))
26     zip_var.set(config.get("zip_default",False))
27     def run_backup():
28         path=path_var.get()
29         if not path:
30             messagebox.showerror("Error","Please select a folder!")
31             return
32
33         new_config={
34             "last_source":path_var.get(),
35             "zip_default":zip_var.get()
36         }

```

4.1.

```

37         save_config(new_config)
38
39         success=Backup.backup(path, zip_it=zip_var.get())
40
41         if success:
42             messagebox.showinfo("Success", "Backup completed successfully!")
43         else:
44             messagebox.showerror("Error", "Backup failed! Check the log file for details.")
45
46     tk.Button(root,text="Start backup", command=run_backup).pack(pady=10)
47     root.mainloop()
48
49
50 if __name__=="__main__":
51     start_gui()

```

5. JSON Config manager

```
1  import json
2  from pathlib import Path
3
4  config_file=Path("config.json")
5
6  def load_config():
7      default_config={"last_source":"","zip_default": False}
8
9      if not config_file.exists():
10         return default_config
11
12     with config_file.open("r") as f:
13         return json.load(f)
14
15  def save_config(config_data):
16     with config_file.open("w") as f:
17         json.dump(config_data,f,indent=4)
18
```

5.1

```
File-Backup-Logger > {} config.json > ...
1  {
2  //    "last_source": "C:/Users/glukhava/EUTe Engineers/File-Backup-Logger/original_dir",
3  //    "zip_default": true
4  }
```